Information Security

Mock 1

Academic Year 2026–2027

Duration: 1 h 30

Instructions. Answer all questions. Responses must be *concise* and every claim must be *justified*. Calculators and external material are not permitted.

Part I – Web Privacy and Data Protection

A company offers free web hosting from the host `webhost.example`. Its relevant features are:

- Each customer may publish a personal site, consisting of arbitrary HTML and JavaScript, served from `http://webhost.example/sites/<customer>/`. For example, a customer named Alice publishes at `http://webhost.example/sites/alice/`.
- Customers manage their hosting through a control panel at `http://webhost.example/panel/`. A customer signs in there with a username and password; on success the server issues a session cookie.
- The server accepts connections over both HTTP and HTTPS.

The session cookie has the following attributes:

Attribute	Value
Name	`session`
Value	(random session identifier)
Path	`/`
Domain	`webhost.example`
SameSite	`Strict`

Neither the `Secure` attribute nor the `HttpOnly` attribute is set.

Q1. Bob is signed in to the control panel and is browsing the site over plain HTTP. The attacker Eve passively monitors his network traffic. Can Eve recover the value of Bob's `session` cookie? State your reasoning.

Q2. Alice may publish arbitrary HTML and JavaScript at `http://webhost.example/sites/alice/`, but holds no other access: she cannot alter the hosting software and cannot read its logs. Determine whether Alice can obtain the `session` cookie of a customer who, while signed in to the control panel, visits her site. If she can, describe concretely how she would construct her page to capture it; if she cannot, explain what prevents her.

Q3. The hosting company weighs several changes to the way the cookie is issued.

(a) Suppose the `HttpOnly` attribute is added to the cookie. State, with reasons, how this affects your answers to Questions 1 and 2.

(b) Suppose instead that the `Secure` attribute is added (and `HttpOnly` remains unset). State, with reasons, how this affects your answers to Questions 1 and 2.

Q4. For the two independent scenarios below, return to Alice's situation in Question 2 and disregard the attribute changes considered in Question 3.

(a) The control panel is moved to the host `panel.webhost.example` and the cookie's `Domain` attribute is set to `panel.webhost.example`, while customer sites remain at `webhost.example/sites/...`. Does your answer to Question 2 still hold? Explain.

(b) Alternatively, the cookie keeps `Domain=webhost.example`, but customer sites are moved to per-customer subdomains, so that Alice's site is now at `alice.webhost.example`. Does your answer to Question 2 still hold? Explain.

Part II – Cryptography

Scenario. Alice is assembling cryptographic building blocks for a project. The adversary Eve can observe all network traffic, store the messages she sees, and on occasion inject or replace messages of her own.

A. Key establishment and message integrity

Q1. Alice and Bob run the Diffie–Hellman protocol over the prime $p = 23$ with generator $g = 5$. Alice's secret exponent is $a = 6$ and Bob's secret exponent is $b = 15$.

(a) Compute the public value each party sends. Show the intermediate steps.

(b) Compute the shared secret, and verify that both parties obtain the same value.

Q2. Alice and Bob already share a secret key K, and Eve may alter messages in transit. Describe a mechanism that allows Bob to verify that a message he receives has not been modified, using their shared key. Give the sequence of steps.

B. Analysis of a coin-toss protocol

Alice and Bob want to decide, by a fair coin toss carried out over an insecure network with no trusted third party, who moves first in an online game. Each party contributes one random bit, and the outcome is the exclusive-or of the two bits (say, 0 means Alice moves first and 1 means Bob moves first). The design goals are that neither party can bias the outcome, and that neither can change its bit after learning the other's.

Q3. Consider the naive protocol in which the two parties simply send their bits, in the clear, one after the other. Identify the opportunities for cheating that this protocol permits.

Q4. To address this, Alice proposes the following scheme:

- Each party encrypts its own bit under a secret key it selects (Alice uses K_A, Bob uses K_B) and sends the resulting ciphertext.
- Once both ciphertexts have been exchanged, each party reveals its key, and the other decrypts and learns the bit.

(a) Identify any remaining opportunities for cheating, and justify your answer.

(b) Could the symmetric cipher be replaced by a public-key encryption scheme? Would such a substitution make the protocol secure? Justify your answer.

Q5. Alice and Bob now consider using a cryptographic hash function H.

(a) Design a protocol, based on H, that meets the two design goals stated above. Specify the messages exchanged and the verification each party performs.

(b) Define the *collision-resistance* property of a cryptographic hash function. What happens to your protocol if H fails to be collision-resistant? Explain.

(c) Define the *preimage-resistance* property of a cryptographic hash function. What happens to your protocol if H fails to be preimage-resistant? Explain.

— End of questions —